Jake DeFord

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WORK EXPERIENCE

Lumetrics, Inc. **April 2024 – Present**
Software Engineer *Rochester, NY*

- Next-Gen OptiGauge DAQ Platform: Architected a complete hardware-consolidation strategy using Xilinx Artix-7 FPGAs, eliminating a costly and high-latency legacy stack comprised of an SBC and an NI-DAQmx card.
- Engineered a 10 Msps data acquisition fabric in Verilog entirely from scratch, utilizing exclusively open-source core modules and entirely bypassing proprietary vendor IP blocks.
- Designed and implemented a proprietary binary payload protocol over UDP from scratch, incorporating low-overhead packet framing, sequence tracking, and hardware CRCs for zero-loss telemetry ingestion.
- Developed a complete, custom library of bare-metal communication cores from scratch—including SPI, I2C, UART, RS-232, and RS-485 controllers—to achieve full deterministic control over peripheral systems.
- Championed a decoupled architectural model, streaming raw telemetry directly from hardware to client applications via a line-rate Ethernet stack to isolate metrology DSP from the collection plane.
- Created Python-based hardware verification suites to validate multi-board behavior, and optimized Vivado/xelab synthesis pipelines via multithreading and IP caching.
- Instrument Monitoring & Infrastructure Platform: Designed a real-time instrument telemetry platform using Django (DRF), React, and WebSockets, backed by MySQL to handle complex, high-frequency measurement records.
- Maintained self-hosted development infrastructure using Docker and Gitea, deploying automated GitHub Actions workflows to enforce 90%+ test coverage gates via pytest and Playwright.

Mastodon Design **February 2022 – March 2024**
Software Engineer I *Rochester, NY*

- Recovered $60,000 in hardware assets over two months by leading a troubleshooting initiative that identified and repaired 32 failing boards.
- Engineered a cost-effective, drop-in hardware replacement for expensive commercial test equipment, featuring Ethernet connectivity and support for SPI, I2C, GPIO, ADC, and PWM.
- Developed Python-based validation software and drivers to aggregate multi-board test results, streamlining the hardware verification lifecycle.

EDUCATION

Rochester Institute of Technology **May 2022**
BS, Computer Engineering *Rochester, NY*

- Dean's List Fall 2020

SKILLS & INTERESTS

- **FPGA / RTL**: Verilog; Vivado; AXI/AXIS; SPI; I2C; UART; UDP/Ethernet; DAC/ADC; xelab simulation; Verible; openFPGALoader
- **Full-Stack**: Django / DRF; React; MySQL; Docker; WebSockets; REST APIs
- **CI / CD**: Git Actions; pytest; Playwright / Jest; Ruff; pip-audit; Docker Buildx; private container & PyPI registry
- **Languages**: Python; System Verilog
- **OS / Tools**: Linux; Git